

What English-tuned ablation misses in Slovene

Framing

Abliteration removes refusal behaviour from language models by orthogonalising weight matrices against a "refusal direction" extracted from English harmful-vs-harmless contrasts [1]. The procedure was introduced by Arditì et al. [1] and automated by tools such as Heretic, which uses TPE search over layer, position, direction index, and LoRA rank to find ablation configurations that minimise an English refusal objective while bounding KL divergence from the original model. The question motivating this study is whether that English-centred objective leaves Slovene refusal behaviour uncontrolled. If the English refusal direction is universal across languages, as Wang et al. [2] report for safety-aligned models evaluated across 14 languages, then ablation tuned on English should transfer to Slovene. If instead Slovene refusal relies partly on representations that the English objective does not observe, a gap will remain. The practical consequence is that a model ablated to comply in English might still refuse in Slovene, or might lose Slovene capabilities that the English-only KL constraint does not protect.

Several concurrent lines of work inform this question. BabelSteering [4] and Yoon et al. [5] show that safety alignment imposes disparate costs across languages, with non-English users bearing higher utility loss. Aziz et al. [6] diagnose the breakdown point for low-resource safety as an action failure rather than a representation failure: the model encodes harmfulness correctly but fails to convert the representation into refusal. Wu et al. [7] formalise this as a disentangled safety hypothesis, separating a recognition axis from an execution axis. On the ablation side, Fafula [8] documents off-target effects of refusal removal across model families, and Young [14] and Petrov [15] compare ablation methods across architectures. Upadhyaya and Sikdar [9] analyse how safety and language identity become entangled, while Hawkins et al. [10] show that benign multilingual fine-tuning has heterogeneous safety impacts. Labunets [11] demonstrates that refusal geometry reflects refusal training diversity. On quantisation effects relevant to our NF4 setup, Marchisio et al. [12] and Chimoto et al. [13] show that quantisation calibration language matters for multilingual performance. Tang et al. [16] identify language-specific neurons, and Ghussin et al. [17] propose principled layer selection for multilingual steering. Frank [18] argues that refusal-based alignment evaluation conflates detection with routing.

We study this question in a controlled bilingual setting using two sibling 12B models from the Gemma-3 family: google/gemma-3-12b-it (the English-centric reference) and cjvt/GaMS3-12B-Instruct (a Slovene continual-pretraining and SFT variant of the same architecture). Both are loaded in bnb_4bit NF4 quantisation throughout. The study proceeds in three iterations. Iteration 1 runs matched Heretic ablation on both models, screens for cross-language refusal-direction transfer, and builds the shared bilingual evaluation dataset. Iteration 2 uses that dataset to measure the surrogacy gap (what English outcomes fail to predict about Slovene), test causal mechanisms, and evaluate safety and utility on held-out benchmarks. Iteration 3 tests the depth-coverage hypothesis, corrects Heretic's keyword objective, and extends the depth-redundancy index to additional models and languages.

[Figure fig_behaviour_summary: generated in the report phase, not in this draft]

Iteration 1

Experiment 1: Matched Heretic runs

Goal. Run identical Heretic TPE searches on both models under the same seed, select an ablated checkpoint for each, and compare the two models' responses to the same edits.

Setup. Both models were searched over 116 complete trials (seed 20260923) using Heretic (SHA 3521f864) with the default configuration: `orthogonalize_direction = true`, `row_normalization = 'full'`, rank-3 LoRA. The English keyword refusal rate (out of 100 harmful prompts) and mean KL divergence served as Heretic's two objectives. Selection followed a frozen protocol: minimise KL subject to refusals ≤ 10 ; if no trial met this primary criterion, fall back to minimising refusals subject to $KL \leq 1.0$. Both models fell back to the secondary rule.

Selected trials. GaMS3 trial 88 reached 16/100 English refusals at KL 0.175. Gemma trial 96 reached 69/100 English refusals at KL 0.024. The Gemma model refused far more often than GaMS3 under identical ablation parameters, and no Gemma trial achieved $\leq 50/100$ refusals. The $13.4\times$ figure is the ratio of medians of refusal drop per unit KL across the 60 shared startup edits (GaMS3 1449 vs Gemma 108 refusals/nat), not a ratio of median refusal counts [Correction, iter 3: clarified definition from evaluation audit]. In all 60 startup edits, Gemma refused at least as often as GaMS3 (60/60, median paired difference +25). In a KL-matched subset ($n=30$, GaMS3 KL median 0.0101, Gemma 0.0098), GaMS3 median refusal was 65/100 versus Gemma 99/100 [Correction, iter 3: added from evaluation audit].

Behaviour on the selected trials. The table below shows keyword refusal rates (out of 100 harmful prompts) and FLORES + NLL for each checkpoint.

Model	Checkpoint	EN refusals	SL refusals	KL mean	FLORES NLL	EN FLORES NLL SL
GaMS3	original	98	97	0.000	3.332	2.570
GaMS3	own edit (trial 88)	16	10	0.175	3.334	2.566
GaMS3	swap (Gemma trial 96)	25	25	0.046	3.336	2.576
Gemma	original	100	97	0.000	4.018	3.702
Gemma	own edit (trial 96)	69	90	0.024	4.012	3.703
Gemma	swap (GaMS3 trial 88)	53	25	0.254	4.011	3.712

[Correction, iter 3: The Gemma rows above are recomputed from the saved evaluation files. The previous draft reported Gemma orig SL = 100 (actual: 97), Gemma own edit SL = 85 (actual: 90), Gemma swap EN/SL = 91/95 at KL 0.293 (actual: 53/25 at KL 0.254), and FLORES values of 3.930/3.427 etc. that are untraceable in iteration-1 files. The corrected FLORES values come from the 40-sentence devtest means stored in the evaluation JSON files.]

GaMS3's own edit reduced English refusals from 98 to 16 and Slovene refusals from 97 to 10. Slovene refusal dropped even though Heretic optimised only on English, and FLORES NLL changed by at most 0.01. For Gemma, the edit reduced English refusals from 100 to 69 but Slovene refusals only from 100 to 90 (keyword count). [Correction, iter 3: the previous draft reported Gemma own-edit SL as 85; the evaluation audit finds 90.]

The swap results are informative. GaMS3 under Gemma's trial-96 edit showed 25/25 (EN/SL) refusals; Gemma under GaMS3's trial-88 edit showed 53/25 (EN/SL) at KL 0.254 [Correction, iter 3: the previous draft reported 91/95 at KL 0.293; the corrected values from evaluation audit show GaMS3's broader, stronger parameters suppress Gemma's SLOVENE refusal far more than English

(SL 97→25 vs EN 100→53), at roughly $10\times$ the KL of Gemma's own edit. This supports the depth/dose reading: a stronger edit of the same base largely transfers to Slovene.].

Direction geometry. Across layers 24 to 47, the mean cosine between the two models' diff-in-means refusal directions was 0.57; across all layers it was 0.63. The cosine between their harmless-token directions was 0.98. Random directions gave a mean $|\cos|$ of 0.014. The refusal directions are more aligned than chance but far from parallel; the harmless directions are near-identical, consistent with the shared tokenizer and architecture.

A3 screen (paired startup edits). The Spearman correlation of refusal counts between the two models across 60 paired edits was 0.78 [0.63, 0.88]. KL divergence correlated at 0.97. However, the incremental variance explained by the sibling model's outcome (dR2_sibling) was near zero for both directions of prediction, meaning the two models' refusal counts are correlated through shared parameters but one model's outcome does not add predictive value over those parameters. The screen verdict was "intermediate" (raw rho 0.781 is below the 0.9 threshold for "shared via strength only") [Correction, iter 3: previous draft said "shared via strength only"; the evaluation audit restores the correct frozen reading].

Sanity checks. All 60 startup edits used identical parameter draws (verified). FLORES NLL shifts were ≤ 0.01 for both models. Slovene marker validation on the selected GaMS3 edit showed accuracy 0.75 and kappa 0.48 (fair agreement between keyword markers and the executor's hand labels on 40 items) [Correction, iter 3: previous draft described these as "LLM-judge labels"; they were executor labels, not native-speaker labels].

Dataset construction

Goal. Build a frozen bilingual (EN/SL) evaluation dataset that spans safety, utility, and mechanistic probing, using matched translation quality controls.

Result. The dataset contains 48,696 rows across 10 blocks (S1 to S7 families): S1 (Heretic direction/KL calibration, 2000 rows), S2 (semantic harmful/harmless pairs, 1664), S3 screen prompts (540 prompts + 640 utility items), S4 (StrongReject harmful/harmless twins, 1028), S5 (RefusEU [3], 2800), S5X (RefusEU cross-translations, 1400), S6 (XSTest over-refusal, 900), S7 (Slovenian LLM eval tasks including ARC, BoolQ, HellaSwag, 35,700; FLORES + devtest, 2024). Protocol hash: dc33bde4.

Translation. Slovene translations were produced by Gemini-2.5-flash (temperature 0) as the primary translator, with NLLB-200-distilled-1.3B as fallback for 8 rows where Gemini refused. Mean back-translation chrF was 81.25; 3 rows fell below chrF 40. Mean LaBSE cross-lingual similarity was 0.856. Quality control was automated only (back-translation chrF, LaBSE, GlotLID); no native-speaker review was conducted. S5 (official RefusEU) EN and SL rows that share a row_id are NOT translations (0 of 1,400 pairs grade T); paired cross-language claims therefore use S5X only [Correction, iter 3: provenance note added from evaluation audit].

Splits. Items are assigned to halves A and B by SHA-1 hash of their semantic ID. This split is used throughout iteration 2 for placebo-controlled surrogacy analyses: English traits computed on half A predict English (placebo) and Slovene (test) on half B.

Experiment 3: A1 screen, EN vs. SL refusal-direction transfer

Goal. Screen whether the English-extracted refusal direction transfers to Slovene, using a 2×2 matrix (source language $\times$ evaluation language), increment tests, and multiple controls. This was a screen, not a confirmatory test.

Setup. For each model, the experiment extracted refusal directions from half-A items using diff-in-means on harmful vs. harmless activations, separately for English and Slovene. It then applied 15 ablation conditions per model (activation steering at inference) to half-B items and measured refusal rates using keyword markers and an LLM judge. The judge was gpt-4.1 for all 1,596 outputs [Correction, iter 3: previous draft described a Qwen3-14B second judge for part of the outputs; the evaluation audit finds gpt-4.1 was the sole judge for exp3].

Precision deviations. Several deviations from the original plan were recorded. The Gemma candidate grid was cut to every-second layer and positions {-1, -5} for GPU time constraints. Translation used a system/user wrapper prompt because bare Gemini prompts elicited refusal. NLLB served as fallback for 22/1070 translation rows. Winsorization ($q = 0.995$) was applied to directions because one massive-activation dimension dominated the raw diff-in-means.

Validity gates. The R validity gate (refusal readout Spearman correlation with judged refusal across 16 conditions) passed for Gemma (Spearman 0.91, AUROC 0.96) but failed for GaMS3 (Spearman 0.51, AUROC 0.70). The GaMS3 failure means the keyword-based refusal readout did not track actual refusal behaviour well enough for the screen's quantitative tests to be trusted. Per the frozen protocol (rule F6), the screen's primary outcome falls to the judged refusal rate for GaMS3.

Judged 2×2 table (gpt-4.1). [Correction, iter 3: added from evaluation audit]

Model	Condition	EN	SL
GaMS3	C0 no-op	0.90	0.93
GaMS3	C1 d_EN	0.46	0.34
GaMS3	C2 d_SL	0.07	0.10
Gemma	C0 no-op	0.83	1.00
Gemma	C1 d_EN	0.07	0.85
Gemma	C2 d_SL	0.29	0.93

[Correction, iter 3: the previous draft reported Gemma EN baseline as "99%"; the gpt-4.1 judged rate is 0.83. The d_EN ablation took Gemma EN from 0.83 to 0.07 while SL stayed at 0.85; d_SL took EN to 0.29 and SL to 0.93. The difference-in-differences for Gemma under d_EN is +0.61 [0.41, 0.78]: the Slovene drop was 0.61 units smaller than the English drop.]

GaMS3 results. Because the R validity gate failed, quantitative screen results for GaMS3 are exploratory. d_EN reduced GaMS3 from 0.90/0.93 (EN/SL) to 0.46/0.34, showing near-parallel reduction across languages. The DiD was -0.15 [-0.32, 0.02], not significantly different from zero.

Gemma results. The Gemma validity gate passed. Applying the English refusal direction d_EN removed most English refusal (0.83 to 0.07) but Slovene refusal survived at 0.85. This is the first signal that the English direction misses Slovene refusal in Gemma.

Screen verdict. WEAK. The predicted cross-language transfer contrast was reversed for Gemma. For GaMS3, the validity gate failure prevented a quantitative verdict. The screen provided the initial signal that English ablation may miss Slovene refusal, motivating the confirmatory analyses in iteration 2.

Matched-efficacy grid. Conditions rescaled so that English refusal dropped by equal amounts showed that Slovene refusal systematically persisted more than English refusal at every matched English decrement. The Heretic bridge mean gap was GaMS3 0.109 [0.06, 0.17] and Gemma 0.235 [0.16, 0.30] [Correction, iter 3: numbers from evaluation audit].

Addition dose-response. Increasing the steering coefficient (scaling up the English refusal direction) reduced both English and Slovene refusal, but Slovene always lagged behind English: at every dose level, Slovene retained more refusal than English.

Iteration 2

Iteration 2 built on the screen results from iteration 1 by running five pre-registered experiments that measured the surrogacy gap, tested causal mechanisms, and evaluated safety and utility on held-out data.

Experiment 5: Utility and mechanism

Goal. Evaluate whether the selected ablation edits preserve utility across both languages and characterise the internal geometry of the harm representation before and after editing.

T1: Harmful behaviour (S4 harmful items, 257 per cell)

The table below shows judged refusal rates on held-out harmful items (S4, 257 items per model $\times$ checkpoint $\times$ language). The gpt-4.1 judge labelled a random $\sim 52\%$ subsample; remaining items used a second-family free judge (Nemotron-3-ultra-550b); markers show the Heretic keyword rule on all 257.

Model	Ckpt	Lang	Refused % [95% CI]	Complied % [95% CI]
GaMS3	orig	EN	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]
GaMS3	orig	SL	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]
GaMS3	edit	EN	5.6 [2.4, 9.5]	92.9 [88.1, 96.8]
GaMS3	edit	SL	6.7 [3.0, 11.1]	89.6 [84.4, 94.1]
Gemma	orig	EN	99.3 [97.8, 100.0]	0.7 [0.0, 2.2]
Gemma	orig	SL	99.2 [97.6, 100.0]	0.8 [0.0, 2.4]
Gemma	edit	EN	70.3 [61.9, 78.0]	24.6 [16.9, 32.2]
Gemma	edit	SL	95.3 [92.0, 98.7]	4.0 [1.3, 7.3]

For GaMS3, the edit reduced refusal from 100% to $\sim 6\%$ in both languages, with no meaningful EN-SL gap. For Gemma, the edit reduced English refusal from 99% to 70% but left Slovene refusal at 95%. This 25-percentage-point gap (70.3% vs. 95.3%) confirms the pattern seen in the iteration-1 screen: the English-tuned Heretic edit substantially missed Slovene refusal in Gemma.

Judge agreement: second judge vs. gpt-4.1 on a stratified sample of 200 gave 6-way kappa 0.765, refused-vs-not kappa 0.828, raw agreement 0.885.

T2: KL divergence on harmless twins (S4, 128-token generations)

Model	Ckpt	Lang	KL1 median [95% CI]	KL32 median [95% CI]	Benign refusal %	Malformed %	Empty %
GaMS3	orig	EN	0 (ref)	0 (ref)	0.0	0.0	0.0
GaMS3	orig	SL	0 (ref)	0 (ref)	0.0	0.0	0.0
GaMS3	edit	EN	0.0441 [0.0323, 0.0530]	0.0228 [0.0196, 0.0250]	0.0	0.0	0.0
GaMS3	edit	SL	0.0174 [0.0152, 0.0211]	0.0068 [0.0061, 0.0086]	0.0	0.0	0.0
Gemma	orig	EN	0 (ref)	0 (ref)	0.0	0.0	0.0

Gemma orig	SL	0 (ref)	0 (ref)	9.5	0.4	0.0
Gemma edit	EN	0.0000 [0.0000, 0.0000]	0.0150 [0.0125, 0.0171]	0.0	0.0	0.0
Gemma edit	SL	0.0113 [0.0083, 0.0163]	0.0037 [0.0030, 0.0046]	0.0	0.4	0.0

KL1 values ranged from 0.011 to 0.044. Gemma EN edit showed KL1 median 0.0000 (at the first token the edit barely changed harmless outputs). Output validity was clean across all cells.

T3: Utility (0-shot, 250 items per task per language)

Model	Lang	Macro original	Macro edited	Change [95% CI]	Holm p
GaMS3	EN	72.5	72.7	+0.13 [-0.53, +0.80]	1.000
GaMS3	SL	64.9	65.1	+0.20 [-0.73, +1.13]	1.000
Gemma	EN	71.5	71.6	+0.13 [-0.53, +0.80]	1.000
Gemma	SL	62.1	62.1	+0.00 [-0.60, +0.60]	1.000

All macro changes were ≤ 0.20 points. All Holm-corrected p-values were 1.000 over the 4 macro comparisons. Individual task changes ranged from -1.20 to +1.60 points; none reached significance. The edits did not measurably alter utility in either language for either model.

T4: FLORES+ per-token NLL

Model	Lang	NLL original	NLL edited	Change [95% CI]
GaMS3	EN	3.3072	3.3068	-0.0004 [-0.0025, +0.0016]
GaMS3	SL	2.4524	2.4526	+0.0002 [-0.0012, +0.0016]
Gemma	EN	3.9194	3.9174	-0.0020 [-0.0031, -0.0008]
Gemma	SL	3.4289	3.4293	+0.0004 [-0.0007, +0.0016]

NLL shifts were ≤ 0.002 for all cells. The Gemma English NLL decrease (-0.002) was statistically below zero but negligible in magnitude.

T5: Bilingual harm geometry (original models, primary site)

Model	Layer	cos(d_EN, d_SL)	Corrected CV AUROC	DiM EN/SL	S4 frozen DiM EN/SL	cos(lang dir, d_EN)	Shuffled AUROC
GaMS3	34	0.832	0.890	0.963 / 0.945	0.998 / 0.997	0.035	0.490
Gemma	20	0.919	0.979	0.979 / 0.941	0.997 / 0.996	0.109	0.515

Cross-validation AUROC on S3 development items exceeded 0.94 for both models in both languages. Cross-language transfer on held-out S4 items exceeded 0.995 in both directions for both models. The harm direction is shared across languages. The language-identity direction is near-orthogonal to d_EN (cos 0.035 and 0.109). Shuffled-label AUROC was at chance (0.49 to 0.52), confirming the directions are not artifacts of data leakage.

Both models encode harmfulness similarly across EN and SL in their representations. The harm direction is shared; the language direction is separate. For Gemma, the behavioural gap (70.3% vs. 95.3% residual refusal) cannot be attributed to a representational difference: the model knows the prompt is harmful in both languages, but acts on that knowledge differently.

T6: Frozen vs. refit probes (S4, primary site)

Model	Lang	Probe	Frozen orig AUROC	Frozen edit-orig [95% CI]	Refit edit-orig [95% CI]	Class
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GaMS3	EN	dim	0.998	-0.401 [-0.434, -0.368]	-0.014 [-0.021, -0.007]	INFO-LOSS
GaMS3	EN	lr	0.999	-0.596 [-0.623, -0.570]	-0.000 [-0.001, 0.001]	DRIFT
GaMS3	SL	dim	0.997	-0.576 [-0.611, -0.542]	-0.043 [-0.056, -0.030]	INFO-LOSS
GaMS3	SL	lr	0.996	-0.661 [-0.691, -0.635]	-0.000 [-0.002, 0.002]	DRIFT
Gemma	EN	dim	0.997	-0.000 [-0.001, 0.000]	-0.000 [-0.001, 0.000]	PRESERVED
Gemma	EN	lr	0.996	-0.001 [-0.002, 0.000]	-0.000 [-0.000, 0.000]	PRESERVED
Gemma	SL	dim	0.996	-0.000 [-0.001, 0.000]	-0.000 [-0.001, 0.000]	PRESERVED
Gemma	SL	lr	0.989	-0.002 [-0.003, -0.001]	0.000 [-0.000, 0.000]	DRIFT

All-layer profile (minimum frozen-probe AUROC on edited activations):

Model Lang Min frozen-edit AUROC (layer) Refit AUROC there

GaMS3	EN	0.201 (L27)	0.993
GaMS3	SL	0.196 (L27)	0.987
Gemma	EN	0.507 (L4)	0.709
Gemma	SL	0.492 (L6)	0.740

For GaMS3, the frozen probe collapsed after editing: AUROC dropped to 0.201 at the worst layer, but a refit probe recovered to 0.993. The edit moved the harm signal away from its original subspace (INFO-LOSS on dim, DRIFT on logistic), but the information persisted in a different location.

For Gemma, the frozen probe was preserved: AUROC dropped by at most 0.002. The edit did not move the harm representation.

The contrast is informative. GaMS3's edit reached low refusal in both languages by destroying the harm signal at the probe layer. Gemma's edit left the harm signal intact, consistent with its failure to suppress Slovene refusal: the model still encodes harmfulness, but the readout mechanism was partially disrupted for English while the Slovene readout remained functional.

T7: Drift geometry

For GaMS3, Cohen's d on the frozen harm axis dropped from 6.20 to 0.30 (EN) and from 5.61 to -0.18 (SL). The share of mean-difference energy on the frozen axis dropped from 0.71 to 0.002 (EN). The cosine between the pre- and post-edit refit axes was 0.274 (EN) and 0.188 (SL), confirming the edit rotated the harm axis. Collapse onset was at layer 22.

For Gemma, Cohen's d dropped from 5.29 to 4.94 (EN) and from 3.76 to 3.50 (SL). The frozen axis retained most of its energy share (0.83 → 0.80). The cosine between refit axes was 0.981. The edit barely moved the harm signal.

T8: Readout validity gate (S4 judged items)

Model	Lang	R_seq item AUROC	R_seq 16-cell Spearman	R_seq pass	R1 item AUROC	R1 pass	Primary trait
GaMS3	EN	0.989	0.867	True	0.979	False	R_seq
GaMS3	SL	0.995	0.899	True	0.963	False	R_seq
Gemma	EN	0.804	0.861	False	0.897	False	JUDGE_ONLY
Gemma	SL	0.920	0.856	True	0.968	True	R_seq

GaMS3 R_seq passed in both languages. Gemma R_seq passed only for SL; EN was classified as

JUDGE_ONLY, meaning the judge labels should be trusted over the R_seq score for Gemma EN cells. R1 did not pass for GaMS3 in either language.

T9: Criterion shift vs. evidence loss

Both models showed EVIDENCE_LOSS under both R_seq and R1 readouts: the slope of the decision function decreased after editing, indicating the internal signal weakened rather than the threshold shifting. For GaMS3, the R_seq slope ratio was 0.36 (EN) and 0.42 (SL). For Gemma, the R_seq slope ratio was 0.61 (EN) and 0.48 (SL).

T10: r_prior geometry (original models)

The false-refusal direction r_prior (constructed from prompts judged harmless that the model refuses, orthogonalised to d_EN) was defined for both models. For GaMS3, 36 refusals (15 EN, 21 SL) out of 275 r_prior generations; cos(r_prior, d_EN) before orthogonalisation was 0.824 (pooled). For Gemma, 55 refusals (8 EN, 47 SL) out of 273; cos(r_prior, d_EN) was 0.631. The within-SL cos(r_prior, lang_dir) was -0.038 (GaMS3) and -0.139 (Gemma), confirming r_prior is not simply the language direction. r_prior is, however, substantially language-dominated: cos with the language axis is 0.65, and cos with its shuffled twin is 0.89 [Correction, iter 3: added from evaluation audit].

T12: Edit-strength dose-response (exploratory)

Model	Scale f	Marker refusal EN %	Marker refusal SL %	KL1 median EN	KL1 median SL
GaMS3	0.0	95.8	100.0	0.0000	0.0000
GaMS3	0.5	75.0	68.8	0.0088	0.0037
GaMS3	1.0	0.0	0.0	0.0405	0.0177
GaMS3	1.5	0.0	2.1	0.1610	0.1167
Gemma	0.0	93.8	100.0	0.0000	0.0000
Gemma	0.5	60.4	95.8	0.0000	0.0046
Gemma	1.0	27.1	91.7	0.0000	0.0103
Gemma	1.5	22.9	68.8	0.0000	0.0303
Gemma	2.0	12.5	31.2	0.0000	0.0924
Gemma	3.0	4.2	8.3	0.0000	0.8228

For GaMS3, scaling the LoRA edit by $f = 0.5$ brought both languages to 68 to 75% refusal; at $f = 1.0$ both reached 0%. EN and SL refusal declined in parallel. For Gemma, at $f = 1.0$ (the core edit) EN refusal was 27% while SL was 92%. Reaching SL refusal below 10% required $f = 3.0$, at which point KL divergence rose to 0.82 (SL), well above Heretic's operational range. This dose-response confirms that the EN-SL gap in Gemma is not an artifact of edit strength: at every dose level, SL refusal exceeded EN refusal.

[Figure fig_dose_response: generated in the report phase, not in this draft]

T13: GaMS3 base model diagnostic (pre-instruction, descriptive)

The uninstructed cjvt/GaMS3-12B base model (before SFT) was tested with a Question/Answer format at layer 34. Frozen S3→S4 diff-in-means AUROC was 0.817 (EN) and 0.773 (SL). Grouped-CV S4 AUROC reached 0.988 (EN) and 0.958 (SL). The cosine between the base model's EN and SL harm directions was 0.781; the cosine between the base model's d_EN and the instruction-tuned model's d_EN was 0.305. Out of 20 EN harmful continuations, 14 complied and 5 refused; out of 20 SL harmful continuations, 13 complied and 5 refused. The base model already encodes a harm axis

that partially survives instruction tuning, but the low cosine with the instruct model's direction (0.305) indicates instruction tuning rotated the harm representation.

Experiment 4: C1 behaviour, RefusEU evaluation

Goal. Evaluate all checkpoints (originals, edits, community ablation reference) on the held-out RefusEU multilingual refusal benchmark [3] in both EN and SL, measuring refusal rate, attack success rate, over-refusal, and response validity.

Judge. The primary judge was a local Qwen3-14B model (agreement with gpt-4.1: kappa 0.83 on 6-way labels, 0.91 on refused-vs-not; tested on a calibration set before deployment). The gpt-4.1 judge was blocked partway through by API budget limits (716 of 3,840 items labelled before the block), so local Qwen3-14B served as the judge for all remaining generations [Correction, iter 3: corrected provenance from evaluation audit; Qwen3-14B was validated against the 716 gpt-4.1 labels before substitution].

Headline RefusEU (S5) refusal rates:

Model	Checkpoint	EN S5 refusal	SL S5 refusal	EN S6 over-refusal	SL S6 over-refusal
GaMS3	original	0.986	0.871	0.100	0.107
GaMS3	edit	0.014	0.000	0.000	0.000
Gemma	original	0.971	0.939	0.087	0.327
Gemma	edit	0.287	0.739	0.013	0.193
Community ref -		0.054	0.114	0.007	0.027

All changes from original to edit were significant (Holm-corrected $p < 1e-14$ for refusal reductions). Invalid rate was 0.000 across all cells.

GaMS3. The edit reduced EN refusal from 0.986 to 0.014 and SL refusal from 0.871 to 0.000. Both languages reached near-zero refusal. Over-refusal dropped from ~ 0.10 to 0.000 in both languages. GaMS3's baseline SL refusal was already lower than its EN refusal (0.871 vs. 0.986), and the edit eliminated refusal in both languages.

Gemma. The edit reduced EN refusal from 0.971 to 0.287 and SL refusal from 0.939 to 0.739. The residual Slovene refusal (0.739) is $2.6 \times$ the residual English refusal (0.287). This is the headline result of the study: an English-tuned ablation edit left most Slovene refusal intact in Gemma while removing most English refusal. Over-refusal dropped in both languages but remained substantial in SL (0.193 vs. 0.013 in EN for the edit). The original Gemma model already had high SL over-refusal (0.327), and the edit reduced it only to 0.193. The PARTIAL class accounts for 0.548 of the Gemma edit's English outputs, meaning more than half of what the keyword counter calls "non-refusal" is caveat-laden partial compliance rather than clean compliance [Correction, iter 3: PARTIAL rate added from evaluation audit].

Community reference ablation. A community Heretic ablation applied to Gemma reached 0.054 EN refusal and 0.114 SL refusal. This more aggressive edit removed most refusal in both languages but had higher FLORES NLL shifts (0.054 EN, 0.111 SL) than the selected trial.

Keyword-proxy miscalibration. The keyword refusal counter, which Heretic uses as its objective, inverts on the edits it selects. On the Gemma edit's English outputs, keyword refusal rate is 0.851 while the Qwen3-14B judged rate is 0.287; the false-positive share is 0.761 (76% of what the keyword calls "refusal" the judge does not), and kappa is -0.04. This inversion arises because the

edit produces PARTIAL responses (warnings plus substantive content) that fire keyword markers but are not full refusals. The keyword counter is near-exact on original models (FP share ≤ 0.03) and structurally unreliable on edited checkpoints [Correction, iter 3: keyword miscalibration table added from evaluation audit].

Paired cross-language analysis (S5X, 100 matched pairs). The paired EN-SL refusal gap depends on the judge and refusal definition. Under the Qwen3-14B strict definition (REFUSED only), the Gemma edit gap is +0.69 [0.60, 0.78]. Under the broad definition (REFUSED + PARTIAL), the gap is +0.23 [0.14, 0.32]. Under the keyword counter, the gap is +0.06 [-0.02, 0.14]. For the community reference, the Qwen3-14B strict gap is +0.12 [0.05, 0.20] [Correction, iter 3: the gap is now stated as a range across judges rather than a single point estimate, per evaluation audit]. The difference-in-differences (edit gap minus original gap) under strict judging was +0.66 [0.55, 0.76].

Experiment 6: GaMS3 surrogacy panel

Goal. Measure the English surrogacy gap for GaMS3: for each ablation trait, how much does an English outcome fail to predict the corresponding Slovene outcome?

Method. 246 Heretic edits (initial startup edits E0 plus TPE-selected edits E_TPE) were generated, activations collected on half-B items, and six traits measured: R (refusal, R_seq readout), R1 (R at first token), Rb (binary refusal), K (KL divergence), N (NLL), M (maximum logit). The primary learner was gradient-boosted trees. The surrogacy gap is defined as $\text{Gap}_t = R2^*(\text{EN}_A \rightarrow \text{EN}_B) - R2^*(\text{EN}_A \rightarrow \text{SL}_B)$, where $R2^*$ is noise-ceiling normalised. The blind share B_t estimates the systematic (edit-determined) portion of the unseen Slovene variation. 160 non-collapsed edits were fitted.

Gap table:

Trait	$R2^*$ placebo	$R2^*$ test	Gap	95% CI	Holm p	B_t [95% CI]
R	0.983	0.968	0.015	[0.006, 0.031]	NA	0.004 [-0.004, 0.009]
R1	0.998	0.956	0.042	[0.027, 0.070]	NA	0.015 [0.002, 0.030]
Rb	0.982	0.954	0.028	[0.012, 0.068]	NA	0.005 [-0.007, 0.013]
K	0.977	0.830	0.147	[0.074, 0.265]	0.000	0.050 [0.001, 0.121]
N	-0.225	0.417	-0.642	[-14.8, 0.98]	0.954	-
M	0.272	0.232	0.040	[-8.37, 5.22]	0.954	-

Interpretation. $\text{Gap}_R = 0.015$ [0.006, 0.031]: English refusal outcomes predicted Slovene refusal almost perfectly ($R2^* = 0.968$), with only 1.5% of the ceiling left unexplained. This gap, while statistically above zero, is small. The blind share B_t for R was 0.004 with a CI spanning zero, meaning the small gap is mostly noise rather than a systematic edit-driven asymmetry. Verdict: CONFIRMATORY. Refusal is visible from English for GaMS3.

$\text{Gap}_K = 0.147$ [0.074, 0.265]: KL divergence showed a substantial surrogacy gap. English KL predicted Slovene KL at $R2^* = 0.830$, leaving 14.7% of the ceiling unexplained. The blind share $B_t = 0.050$ [0.001, 0.121], with the CI excluding zero: part of this gap is systematically determined by the edit, not just noise. Verdict: BLIND. KL divergence has a component in Slovene that English KL does not see. Holm p = 0.000. The SL KL is SMALLER on average (mean-change ratio 0.59); the gap is not that Slovene gets MORE perturbation, but that it gets DIFFERENT

perturbation [Correction, iter 3: direction of KL gap clarified from evaluation audit].

Gap_N and Gap_M showed no reliable signal (NLL and max-logit traits had low reliability in English, so the gap estimate was uninformative).

Carrier regression. The exposure differential D ($\log E_{SL} - \log E_{EN}$) was tested as a carrier of the surrogacy gap. For every trait, $dR2(D \text{ given } S0)$ was near zero or negative. The D carrier FAILED for all traits. The surrogacy gap is not explained by differences in how much the edit perturbs Slovene vs. English token representations on harmless text.

Transfer slopes. The SL/EN transfer slope for R was 0.82 [0.74, 0.90]: when the edit moves English refusal by one unit, Slovene refusal moves by 0.82 units. For K, the slope was 0.39 [0.34, 0.48], much stronger attenuation. For R1, the slope was 0.50 [0.46, 0.55].

Forecast. Conformal prediction intervals at 90% coverage met the 85% threshold on only 50% of trait $\times$ holdout-set combinations. The core edit (trial 88) was out of support for all traits, meaning it lay outside the distribution of TPE-explored edits. This limits the forecast's applicability to the specific selected checkpoint.

Stability checks. Gap estimates were stable across alternative learners, inclusion of collapsed edits, reverse-direction analysis (SL $\rightarrow$ EN), and MT-stable subsets.

Experiment 7: Gemma surrogacy panel

Goal. Measure the English surrogacy gap for Gemma using the same protocol as experiment 6.

Setup. 203 Heretic edits, 163 fitted. The same six traits were measured.

Verdicts [Correction, iter 3: frozen prediction wordings restored to their original form; the previous draft inverted P-a's direction]:

- **P-a (Gap_R $\geq$ 0.10 with 95% LB $>$ 0):** NOT CONFIRMED. The frozen prediction expected a LARGE gap; the observed gap on R1 was 0.054 [0.022, 0.128], which is above zero but below the 0.10 threshold. The gap is small, not large. Label-swap placebo: 15.5% of swaps at least as large as observed.
- **P-b (Gap_Rb $\geq$ 0.10):** UNTESTABLE. The Rb validity gate failed, so the analysis was exploratory. Exploratory estimate: Gap_Rb = 0.018 [0.011, 0.043].
- **P-c (exposure differential carries the gap):** NOT CONFIRMED. $dR2(P \text{ given base})$ and $dR2(b1 \text{ given } P)$ were near zero for both R1 and Rb.
- **P-d (matched-margin gap $\geq$ 0.10):** NOT CONFIRMED. Gap_margin_matched = 0.057 [0.029, 0.130].

Key result. Gap_K = +0.079 (Holm $p = 0.009$): KL divergence showed a surrogacy gap for Gemma as well, though smaller than GaMS3's 0.147. The recomputed transfer slopes are: R1 0.436, R_seq 0.220, lp_ref 0.081, lp_comp 0.849 [Correction, iter 3: slopes recomputed from evaluation audit]. The compliance log-probability slope (0.849) is near 1.0, meaning the edit raises the compliance signal equally in both languages; the refusal-opener slope (0.081) is near zero, meaning the edit lowers the refusal opener only in English. This is what partial depth coverage predicts.

Comparison with GaMS3. The Gemma surrogacy gap on refusal traits was larger than GaMS3's (0.054 vs. 0.015 for R1; overlapping CIs), consistent with the behavioural result that Gemma

retained more Slovene refusal after editing. The KL gap was smaller (0.079 vs. 0.147), but both models showed a significant blind component in KL. Reading: attenuated but predictable [Correction, iter 3: "attenuated but predictable" is the correct characterisation from the frozen predictions; the previous draft omitted this framing].

Experiment 8: Causal test

Goal. Test whether the Slovene refusal that survived English ablation can be causally removed by specific interventions, and identify where in the network the relevant information resides.

Pre-registered hypothesis. Subtracting the false-refusal direction r_{prior} (constructed from prompts judged harmless that the model refuses, following Wang et al. [2]) from the English refusal direction d_{EN} should cut the residual Slovene refusal by at least half, beyond what a random control achieves. The KILL criterion fires if the r_{prior} cut is not significantly larger than the best random control.

Primary result: hypothesis FAILS. Applying d_{EN} alone reduced Gemma's harmful EN refusal from 0.91 to 0.23 but left SL refusal at 0.86. Adding r_{prior} (arm A2: $d_{\text{EN}} + r_{\text{prior}}$) moved SL refusal to 0.83. The r_{prior} cut relative to d_{EN} alone was 0.036 [-0.037, 0.109], not significantly different from zero, and not significantly larger than the best random control (0.074). KILL criteria a and c fired (part 1 failed, and F6 over-refusal test failed). The pre-registered hypothesis was rejected.

Full Gemma activation arms (111 harmful + 111 harmless per language):

Arm	EN harmful	SL harmful	EN harmless	SL harmless
A0 no-op	0.91	0.99	0.05	0.28
A1 d_{EN}	0.23	0.86	0.00	0.19
A2 $d_{\text{EN}} + r_{\text{prior}}$	0.21	0.83	0.00	0.12
A3 r_{prior} alone	0.95	0.98	0.04	0.19
A4 $d_{\text{EN}} + \text{language identity}$	0.23	0.55	0.02	0.08
A7 $d_{\text{EN}} + \text{rand}_1$	0.21	0.85	0.01	0.15
A10 $d_{\text{EN}} + \text{shuffled-label } r_{\text{prior}}$	0.41	0.73	-	-

Several observations from the arm table:

1. **r_{prior} alone (A3)** had almost no effect: SL harmful refusal went from 0.99 to 0.98, EN from 0.91 to 0.95. The false-refusal direction does not carry meaningful refusal-suppression information by itself.
2. **Language-identity direction (A4)** combined with d_{EN} reduced SL harmful refusal from 0.86 to 0.55, the largest single-direction improvement. However, it also raised FLORES NLL by 2.0 (EN) and 1.9 (SL), destroying language modelling quality. This is not a practical intervention.
3. **Random directions (A7, A8, A9)** combined with d_{EN} left SL refusal between 0.74 and 0.85, comparable to d_{EN} alone. Random directions did not systematically help.
4. **Shuffled-label r_{prior} (A10)** combined with d_{EN} reduced SL refusal to 0.73 but also raised EN harmful refusal to 0.41, suggesting the shuffled direction disrupted the English refusal pathway as well.

Exploratory: layer-matched d_EN across all 48 layers (X1). Applying the per-layer English refusal direction at all 48 layers simultaneously reduced SL harmful refusal from 0.86 to 0.10 and EN harmful refusal to 0.03. This is the only intervention that nearly eliminated Slovene refusal, but it required steering at every layer simultaneously, raised FLORES NLL by 0.29 (EN) and 0.52 (SL), and is not equivalent to the single-direction ablation that Heretic performs.

Depth localisation. Applying layer-matched d_EN to different layer bands:

Band	SL harmful	EN harmful	FLORES Δ NLL SL
Layers 1-12	0.82	0.79	0.743
Layers 13-24	0.82	0.45	0.017
Layers 25-36	0.70	0.72	0.082
Layers 37-48	0.99	0.87	-0.003
Layers 1-24	0.55	0.42	0.409
Layers 1-36	0.21	0.38	0.513
All 48	0.23	0.37	0.524

No single band removed Slovene refusal. Layers 13-24 were most effective for English (0.45) but left Slovene at 0.82. Layers 25-36 provided the most Slovene reduction per band (0.70) without large NLL cost. Full Slovene refusal removal required cumulative intervention across at least three bands.

Weight edit repair. The experiment also tested whether augmenting the core Heretic weight edit with additional directions could repair the Slovene gap:

Arm	SL harmful	EN harmful	FLORES Δ NLL SL
W0 core Heretic edit	0.89	0.39	-0.005
W1 core + r_prior weight edit	0.76	0.23	0.009
W3 core + layer-matched d_EN(h)	0.11	0.29	0.546
W4 core + random weight edit	0.47	0.70	0.064

The r_prior weight edit (W1) reduced SL refusal from 0.89 to 0.76 while also reducing EN refusal to 0.23, but the SL residual remained high. The layer-matched weight edit (W3) reached 0.11 SL refusal but at severe NLL cost (+0.55). The random weight edit (W4) reduced SL refusal to 0.47 while raising EN refusal to 0.70, suggesting it disrupted the model rather than targeting Slovene refusal.

Community Heretic reference vs. core edit (S4 held-out, 70 pairs):

Arm	EN harmful	SL harmful	SL harmless
W0 core edit	0.39	0.89	0.07
C0 community edit	0.01	0.26	0.01
C1 community + r_prior	0.01	0.20	0.00

The community edit reached much lower SL refusal (0.26) than the selected trial's edit (0.89), at the cost of slightly higher FLORES NLL (+0.11 SL). Adding r_prior to the community edit reduced SL refusal from 0.26 to 0.20, a modest improvement. The community edit achieved stronger refusal suppression because it was more aggressive overall, not because it targeted Slovene specifically.

GaMS3 descriptive arms. The GaMS3 results were descriptive (not pre-registered for the causal test):

Arm	EN harmful	SL harmful	EN harmless	SL harmless
G0 no-op	0.93	0.95	0.07	0.10
G1 d_EN	0.57	0.50	0.00	0.04

For GaMS3, applying d_EN reduced both EN and SL refusal comparably (from 0.93/0.95 to 0.57/0.50), consistent with the small surrogacy gap found in experiment 6. The EN-SL difference was minor.

Iteration 3

Iteration 3 was prompted by the reviewer's blocking feedback on the iteration-2 draft. Three objections required experimental answers: (1) the headline gap (+0.69 on S5X strict) depends on a single judge, and the keyword proxy that Heretic actually optimises assigns a gap of only +0.06; no number in the draft was stated as a range across judges. (2) The depth localisation from experiment 8 (no single 12-layer band removes Slovene refusal, cumulative three-band steering reaches SL 0.21) suggests coverage matters, but every arm in iteration 2 varied coverage and strength together; there was no matched-energy comparison. (3) The depth-redundancy observation rests on two sibling checkpoints from one architecture; it could be coincidence rather than a measurable quantity.

Iteration 3 therefore ran four experiments and one evaluation audit, each attacking the reviewer's objections from a different angle.

Experiment 9: Gemma coverage × strength factorial

Goal. Test whether depth coverage or total edit energy governs how much Slovene refusal survives, by building weight edits that cross coverage with strength at matched total removal energy.

Setup. Weight edits were constructed using Heretic's operator path (orthogonalize_direction = true, row_normalization = 'full', rank-3 LoRA, per-layer d_EN(h) from experiment 8 directions) on google/gemma-3-12b-it, crossing 10 coverage sets with 4 to 5 per-layer strength levels. Coverage sets included four contiguous 12-layer bands (B1: 1-12, B2: 13-24, B3: 25-36, B4: 37-48), two cumulative sets (C24: 1-24, C36: 1-36), a full-depth set (ALL48), stride-2 (S2) and stride-4 (S4) sets, and the support of the iteration-1 Heretic kernel (K96). Strength was controlled by a per-layer projection coefficient c in {0.25, 0.50, 1.00, 1.50}, with total removal energy E computed analytically for every cell before running. The panel comprised 122 cells scored on 27,784 generations by a local Qwen3-14B judge with the frozen experiment-4 rubric, certified at kappa 0.87 vs gpt-4.1 within edited checkpoints.

PARTIAL counts as compliance throughout. INVALID is excluded from denominators.

Part A: DEV depth-redundancy index. Cumulative-prefix directional ablation (activation-only) on S3 half-A (44 harmful items per language) gave:

Family Lang index_L			
prefix	EN	16	[16, 20]
prefix	SL	20	[20, 28]

suffix EN 24 [24, 24]

suffix SL 24 [24, 28]

The English index (the minimal layer count whose ablation brings refusal below 0.5) is 16; the Slovene index is 20. Slovene needs at least one more 4-layer step of coverage than English. The leave-one-band-out necessity profile shows the 25-36 band is most necessary for Slovene (LOBO necessity +0.39) but not for English (+0.00), indicating Slovene refusal is written more deeply.

[Figure fig_redundancy_index: generated in the report phase, not in this draft]

Part P2: Matched-energy groups. Four groups, each containing a narrow-and-strong and a broad-and-weak cell within the same total removal energy:

Group	E	SL narrow	SL broad	SL contrast [CI]	EN contrast
G1	16.7	0.98	1.00	-0.02 [-0.07, 0.00]	-0.03
G2	19.2	0.63	0.98	-0.34 [-0.49, -0.20]	-0.27
G3	35.6	1.00	0.66	+0.34 [+0.20, +0.49]	+0.67
G4	37.7	0.95	0.73	+0.22 [+0.10, +0.34]	+0.30

The pooled SL contrast (narrow minus broad) was +0.05 [-0.00, 0.10]; the pooled EN contrast was +0.17 [0.09, 0.24]. The difference-of-differences SL-EN was -0.12 [-0.21, -0.03]. **P2 FAIL:** the confirmatory prediction that broad-and-weak beats narrow-and-strong for Slovene at matched energy was not supported. The groups split: G2 favoured narrow (narrow SL 0.63 < broad SL 0.98), while G3 and G4 favoured broad. The direction of the matched-energy contrast depended on WHICH layers carried the energy, not on how many.

Part P1: Coverage regression. The incremental R2 of coverage terms (number of covered layers, depth span) over the base model (log E + EN refusal + static band masses) was dR2 = 0.040 [0.006, 0.136] on 46 cells. The falsifier (dR2 < 0.05) fired. Coverage as a count of layers does not add meaningful predictive power over total energy and placement. **P1 FAIL.**

Part P3: DEV index prediction. The frozen DEV index predicted per-cell residuals out of sample at Spearman 0.78 [0.64, 0.88] for both EN and SL. **P3 PASS** (threshold: Spearman ≥ 0.6). On held-out harm categories, the SL Spearman was 0.72 [0.13, 1.00] (n = 10 cells).

Exploratory: which layers, not how many. The pre-registered exploratory analysis (declared before outcomes were read) revealed that the fraction of layers 13-24 covered by a set predicts the lowest Slovene refusal that set achieves at any strength, at Spearman -0.942 across 10 coverage sets (placebo-verified, p = 0.004). Sets covering all of layers 13-24 (ALL48, C36, K96, B2, C24) all reached SL < 0.20 at sufficient strength. Sets missing layers 13-24 (S4, B3, B1, B4) never reached SL < 0.93 at any strength. The band-mass regression confirmed this: in the SL model (R2 = 0.701), the mass of energy in band 13-24 was the dominant predictor; in the EN model (R2 = 0.811), band 13-24 mass also dominated, but at lower coefficients.

The contiguous-vs-strided comparison at matched energy (33 pairs within 15% log E) showed strided coverage leaves 0.12 [0.07, 0.19] more Slovene refusal while covering more layers on average (-5.5 more), with the effect carried by magnitude rather than by a majority of pairs (17/33).

S5X verified pairs (declared second touch, n = 100). At c = 1.5 with full coverage (ALL48 and

K96), the SL-EN gap was +0.06 [0.00, 0.12], compared to the no-op gap of +0.07 [0.03, 0.12]. The high-dose, full-coverage weight edit closes the gap to near-baseline.

Per-layer write mass (exploratory). English and Slovene refusal write mass spread over the same number of layers (18 to 80% cumulative), but the distribution differs: Slovene's mass sits 60% in band 25-36 vs English's 63% in band 37-48. The prediction that Slovene's mass is spread over MORE layers was FALSE. Instead, it is spread over DIFFERENT layers.

[Figure fig_band_analysis: generated in the report phase, not in this draft]

Experiment 10: GaMS3 coverage test

Goal. Run the identical coverage $\times$ strength factorial and DEV index on GaMS3, the checkpoint whose English-derived edit transferred, to test whether $\text{index_SL} \approx \text{index_EN}$ there.

DEV index: EN = 16 [16, 16], SL = 20 [16, 24]. The Slovene index is one grid step above English, with overlapping CIs. In both models the English index is shallower; the SL lag is common to both checkpoints.

Weight-edit panel (57 cells). The matched-energy groups gave:

Group	E	SL narrow	SL broad	SL contrast
E1	6.9	0.81	0.88	-0.06 [-0.14, 0.00]
E2	13.9	0.55	0.71	-0.16 [-0.23, -0.09]
E3	27.8	0.35	0.44	-0.09 [-0.18, -0.01]

In all three groups, the narrow-and-strong cell (B2 or B3, layers 13-24 or 25-36) left LESS Slovene refusal than the broad-and-weak cell (STR2 or STR4, strided). **PB2 FALSIFIED OPPOSITE:** at matched energy, narrow beats broad for GaMS3, the reverse of what the hypothesis predicted. **PB1 NOT SUPPORTED:** $\text{dR2}(\text{coverage} \mid \text{base}) = 0.002 [0.000, 0.010]$.

Nested R2 (where the variance sits):

Model	R2
logE	0.325
logE + placement (b3 band masses)	0.718
logE + count	0.400
logE + count + span	0.532

Placement (which band the energy sits in) explains 71.8% of the Slovene residual variance, while count (how many layers) explains only 40.0%. The decisive factor is WHERE the energy sits, not HOW MANY layers are covered. This is consistent with experiment 9's exploratory finding: band 13-24 density drives the outcome.

Placement at matched energy AND matched layer count (12 layers, band vs stride-4):

Contrast	SL band	SL stride-4	SL delta
E1_B2 vs STR4	0.83	0.93	-0.10
E2_B2 vs STR4	0.51	0.89	-0.37
E3_B2 vs STR4	0.24	0.81	-0.57

At the same energy, same layer count, only placement differs: a contiguous 12-layer band in layers 13-24 removes far more refusal than every-4th-layer across the full depth. This holds for English as well but the EN gap is smaller, confirming that placement is a general property of edit efficacy, not specific to cross-language transfer.

Cross-model contrast: both models show $\text{index_EN} = 16$, $\text{index_SL} = 20$. The Slovene lag is not what distinguishes these two models under this instrument. Any iteration-1 difference between their Heretic edits (GaMS3 transferred, Gemma did not) must be explained by something other than how redundantly refusal is written across depth.

Experiment 11: Corrected objective

Goal. Test whether replacing Heretic's English keyword refusal counter with a partial-aware score, at equal budget/seed/data/quantisation, makes the search select a kernel that closes the Slovene-English gap.

Setup. A refusal classifier was distilled from existing gpt-4.1 and Qwen3-14B labels (thousands on disk from experiments 4, 5, and 8), using char/word n-gram features, response length, marker hits, and first-64-token embeddings. It was certified at kappa 0.924 (refused-vs-not, within edited checkpoints) against the Qwen3-14B judge. The corrected Heretic run used the same seed (20260923), same 116-trial budget, same NF4 precision, and the same selection rule (lowest KL among trials with ≤ 10 refusals); only the refusal scorer changed.

Selection outcomes. Under the corrected scorer, the primary selection rule fired (not the fallback), selecting trial 7 with 5/100 refusals at KL 0.258. Under the keyword scorer on the same 116 trials, the fallback rule fired, selecting trial 96 with 69/100 keyword-refusals at KL 0.024. The keyword counter fails because it sees 72/100 keyword-refusals on trial 107 (which the judge rates at 10/100 actual refusals) and cannot select it.

Post-hoc reselection (rescoring the original 116 trials with the corrected scorer and re-applying the selection rule) selected trial 98 (classifier) or trial 64 (judge), both with ≤ 10 refusals.

Coverage. The corrected trial 7 has sum_A1 (aligned coverage) = 64.0, versus trial 96's 26.0. The corrected objective bought broader coverage, as predicted by the depth-placement hypothesis.

Headline results (\$5X, 100 verified translation pairs):

Arm	EN refusal	SL refusal	SL-EN gap [95% CI]
A orig	0.950	0.990	+0.04 [+0.01, +0.08]
B keyword t96	0.150	0.830	+0.68 [+0.59, +0.77]
C corrected t7	0.030	0.410	+0.38 [+0.29, +0.47]
D reselected clf t98	0.050	0.350	+0.30 [+0.21, +0.40]
D2 reselected judge t64	0.040	0.450	+0.41 [+0.32, +0.51]
F dose 1.5	0.030	0.360	+0.33 [+0.24, +0.42]
F dose 2.0	0.000	0.000	+0.00 [+0.00, +0.00]
F dose 3.0	0.000	0.000	+0.00 [+0.00, +0.00]

Frozen predictions. Holm-adjusted p-values over the confirmatory family: P1 adjusted 0.411, P2 adjusted 1.0, P3 adjusted 0.0.

- P1 (EN refusal ≤ 0.054): PASS at point estimate (0.03); CI upper bound 0.085 does not exclude 0.054. Holm $p = 0.411$.
- P2 (S5X gap CI upper < 0.35): FAIL. Gap 0.38, CI [0.29, 0.47]. Holm $p = 1.0$.
- P3 (FLORES dNLL $\leq +0.10$): PASS. dNLL +0.003. Holm $p = 0.0$.
- P4' (output validity replacement): SUPPORTED. Invalid EN 0.0, language-consistent EN 1.0, over-refusal EN 0.017 $<$ original 0.133.
- P5 (reselection ratio ≥ 0.5): PASS. Ratio 1.27.
- P6 (corrected A1 $>$ trial 96's): PASS. A1 64.0 $>$ 26.0.
- **P7 falsifier (dose ladder closes gap as well): FIRED.** At equal EN refusal, dose-scaling trial 96 to $c=1.5$ gives a gap of +0.33 [0.24, 0.42], comparable to the corrected edit's +0.38. The value is -0.05 (corrected minus dose at equal EN), which does not reach the 0.15 threshold. This means the gap reduction may be attributable to "more total edit" rather than "better objective."

The corrected objective halved the gap (from +0.68 to +0.38) but did not reach the +0.35 target. The falsifier complicates the causal story: dose-scaling the keyword edit to the same EN refusal level achieves comparable gap reduction, suggesting the coverage gain is partly a side-effect of stronger editing rather than purely better objective design. However, the corrected objective produced this result at KL 0.258, while the dose-1.5 arm needed a gentler KL of 0.073 to get there. The FLORES cost was minimal in both cases (dNLL ≤ 0.005).

Experiment 12: Predictive test across models and languages

Goal. Test whether the depth-redundancy index, measured on development data, predicts out of sample how much of an English-derived weight edit transfers to each unmonitored language across multiple model families.

Setup. Three models: google/gemma-3-12b-it (anchor), Qwen/Qwen3-8B (qwen3), mistralai/Mistral-7B-Instruct-v0.3 (mistral). Four languages: English, Slovene, German, Lithuanian. Per-layer $d_{EN}(h)$ directions extracted by diff-in-means on DEV items. DEV index measured at cumulative prefixes (10%, 25%, 50%, 75%, 100% of depth). Eligibility gate: no-op refusal ≥ 0.60 AND INVALID ≤ 0.20 .

DEV indices:

Model	EN	SL	DE	LT
gemma	0.5	0.75	0.5	0.75
qwen3	0.75	0.75	0.75	0.5
mistral	0.1	0.25	0.1	0.1

Mistral failed the eligibility gate in all languages (no-op refusal 0.07 to 0.53); all four mistral rows and qwen3-LT were excluded. Eight eligible rows remained.

Weight panel (5 cells per model $\times$ language, held-out StrongREJECT items). The Gemma trial-96 adapter applied to gemma showed residual refusal EN 0.60, DE 0.53, LT 0.75, SL 0.92 (exposure to the Slovene residual that motivated the study). For qwen3, the W3 (high-dose) arm brought EN to 0.37 and SL to 0.32, with the gap much smaller than Gemma's.

Frozen predictions: VERDICT FALSIFY.

- **P1** Spearman(index, residual) = -0.009 [-0.131, 0.192] over 21 rows. The index does not predict the cross-model, cross-language residual. Pass = False.
- **P2** concordance 8/12 decided (0.67); binomial p = 0.19. Pass = False.
- **P4** index vs baselines: single-site transfer rho +0.732 beats the index by -0.741. The simple single-site ablation rate predicts better than the frozen index.

The depth-redundancy index, as defined and measured here, does not generalise beyond the two Gemma-3 siblings. It is a two-checkpoint observation, not a predictive instrument.

[Figure fig_cross_model: generated in the report phase, not in this draft]

Evaluation 1: Audit and judge-sensitivity table

Goal. Discharge the reviewer's blocking audit: recompute every draft number from saved files, build the judge-sensitivity table, compile the dead-end ledger, and position against the nearest published work.

Audit headline. 167 draft numbers checked: 137 match, 8 mismatch, 12 misdescribed, 6 untraceable (mismatch rate 5.5%). Two sign-reversed conclusions were found: the iteration-1 swap (see corrections above) and experiment 7 P-a direction (see corrections above). 19 of 99 behavioural claims are judge-sensitive. Placebos passed: 6/6.

Judge-sensitivity table (headline gap as a range). The EN-SL gap for the Gemma edit on S5X depends on judge and definition:

Judge	Definition	Gap	CI
Qwen3-14B	strict (REFUSED only)	+0.69	[0.60, 0.78]
Qwen3-14B	broad (REFUSED + PARTIAL)	+0.23	[0.14, 0.32]
keyword	strict	+0.06	[-0.02, 0.14]

The strict gap ranges from +0.06 (keyword) to +0.69 (Qwen3-14B). The broad gap collapses to +0.23 because 0.548 of the Gemma edit's English outputs are PARTIAL. On GaMS3, all judges agree the gap is near zero (range -0.05 to +0.02).

Dead-end ledger. Twenty items, each with its closing number:

Item	Status	Closing number
iter-1 experiment 2, experiment 4	NOT RUN	empty pods
P2 Sobol sensitivity bands	NOT RUN	cut for time
English-only-vs-full forecast	NOT RUN	50% coverage
language-orthogonalised matched-efficacy arm	NOT RUN	cut for time
Gemini second judge	NOT RUN	OpenRouter 403
exposure differential D	CLOSED	dR2 $\approx$ 0
static geometry b1/b2/b3, LSAR Omega	CLOSED	CV R2 $\leq$ 0
r_prior false-refusal direction	CLOSED	cut 0.036 < best random 0.074

thin-margin rival	CLOSED	margin-matched gap + 0.057
language-identity direction (A4)	WORKS BUT UNUSABLE	FLORES dNLL + 2.0
B3 batching certification	FAILED	non-identical
gpt-4.1 primary judge coverage	PARTIAL	716/3,840 (exp4)
P1 depth-coverage regression (exp9)	FAILED	dR2 = 0.040 < 0.05
P2 matched-energy broad-vs-narrow (exp9)	FAILED	pooled contrast + 0.05, not significant
PB2 matched-energy (exp10, GaMS3)	FALSIFIED OPPOSITE	-0.10
depth-redundancy index generalisation (exp12)	FALSIFIED	Spearman -0.009

Novelty positioning. Wang et al. [2] established direction universality across 14 languages using all-layer activation ablation; Slovene and Gemma-3-12B were not among their models or languages. Arditi et al. [1] showed refusal is mediated by a single direction ablated at all components. This study adds: (a) the English direction does clear Slovene when applied at every depth (consistent with both), but (b) the site-limited Heretic weight edit leaves a gap whose size is judge- and definition-dependent (strict +0.06 to +0.69), and (c) the gap is governed by WHERE the edit energy sits (band 13-24 density, Spearman -0.942) rather than by how many layers are covered.

Pending human review. Five packets totalling 640 items are ready for native-speaker labelling. No native-speaker review has been conducted anywhere in this study; all judges are automated.

What we have learned so far

[Figure fig_refusal_gap: generated in the report phase, not in this draft]

Three iterations have converged on a picture with one firm positive, one partial positive, and three falsified predictions.

Firm positive: band placement governs residual refusal. At matched total removal energy, which 12-layer band carries the energy predicts how much Slovene refusal survives. The band-density Spearman is -0.942 (exp9, placebo-verified, $p = 0.004$): coverage sets including layers 13-24 reach $SL < 0.20$ at sufficient strength; those missing it never reach $SL < 0.93$. This holds for both Gemma and GaMS3: in GaMS3 the nested R2 of placement (0.718) far exceeds count (0.400) (exp10). At matched energy AND matched layer count, a contiguous B2 band (layers 13-24) removes up to 0.57 more Slovene refusal than a stride-4 pattern at the same energy (exp10). The relevant factor is not how many layers the edit touches, but where the energy sits.

Partial positive: corrected objective halves the gap. Replacing Heretic's keyword refusal counter with a partial-aware classifier, at equal budget/seed/data, moved the S5X strict gap from +0.68 to +0.38 (exp11). The corrected selection bought broader aligned coverage ($A1 = 64.0$ vs 26.0). However, the P7 falsifier fired: dose-scaling the keyword edit to the same English refusal level achieved comparable gap reduction (+0.33), meaning the improvement may be "more edit" rather than "better objective." The gap did not reach the pre-registered +0.35 target (observed +0.38, CI [0.29, 0.47]).

Falsified: depth-coverage count does not explain the residual. The coverage regression (exp9 P1, $dR2 = 0.040$) and the matched-energy broad-vs-narrow comparison (exp9 P2, pooled $+0.05$) both failed their confirmatory thresholds. Coverage as a simple layer count does not add meaningful predictive power over total energy and band placement. The depth-coverage hypothesis as originally stated (more layers = less residual) is falsified; the replacement finding is that band identity matters more than coverage extent.

Falsified: matched-energy narrow beats broad in GaMS3. Experiment 10's PB2 was FALSIFIED OPPOSITE: at matched energy, narrow-and-strong cells (B2, B3) left less GaMS3 Slovene refusal than broad-and-weak cells (STR2, STR4), the reverse of the prediction. This makes sense once band identity is the driver: B2 (layers 13-24) concentrates energy in the right band, while a stride pattern dilutes it.

Falsified: the depth-redundancy index does not generalise. Experiment 12 found Spearman -0.009 between the frozen index and the cross-model, cross-language residual. The single-site transfer rate ($\rho +0.732$) was a far better predictor. The index is a two-checkpoint observation, not an instrument.

What survives from iterations 1-2. The behavioural dissociation (GaMS3 transferred, Gemma did not) is robust across judges and datasets. The surrogacy gap on refusal is small for GaMS3 (0.015) and attenuated but predictable for Gemma (0.054). The harm representation is shared across languages (AUROC > 0.995); the behavioural gap is an action failure, not a representation failure, consistent with Aziz et al. [6]. KL divergence has a blind surrogacy gap in both models (0.147, 0.079). Utility is preserved in all cells. The keyword proxy inverts on edited checkpoints ($\kappa -0.04$ on Gemma edit EN).

The headline gap stated as a range. The Gemma edit S5X paired EN-SL gap ranges from $+0.06$ (keyword strict) to $+0.69$ (Qwen3-14B strict), with the broad definition at $+0.23$. On GaMS3, all judges agree the gap is near zero. The PARTIAL class (0.548 of Gemma edit EN outputs) is the wedge: the keyword counter treats partial compliance as refusal, closing the measured gap; the LLM judge treats it as compliance, opening the gap. Neither reading is wrong; they measure different constructs.

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